SQL Assignment 1

Q1.
```
create database library_management_system;
Use library_management_system;
```



Q2. CREATE TABLE authors (

author_id INT PRIMARY KEY,

name VARCHAR(100),

country VARCHAR(50)

);

Q3.

CREATE TABLE books (

book_id INT PRIMARY KEY,

title VARCHAR(150),

author_id INT,

published_year YEAR,

available_copies INT,

FOREIGN KEY (author_id) REFERENCES Authors(author_id)

);

Q4.

CREATE TABLE members (

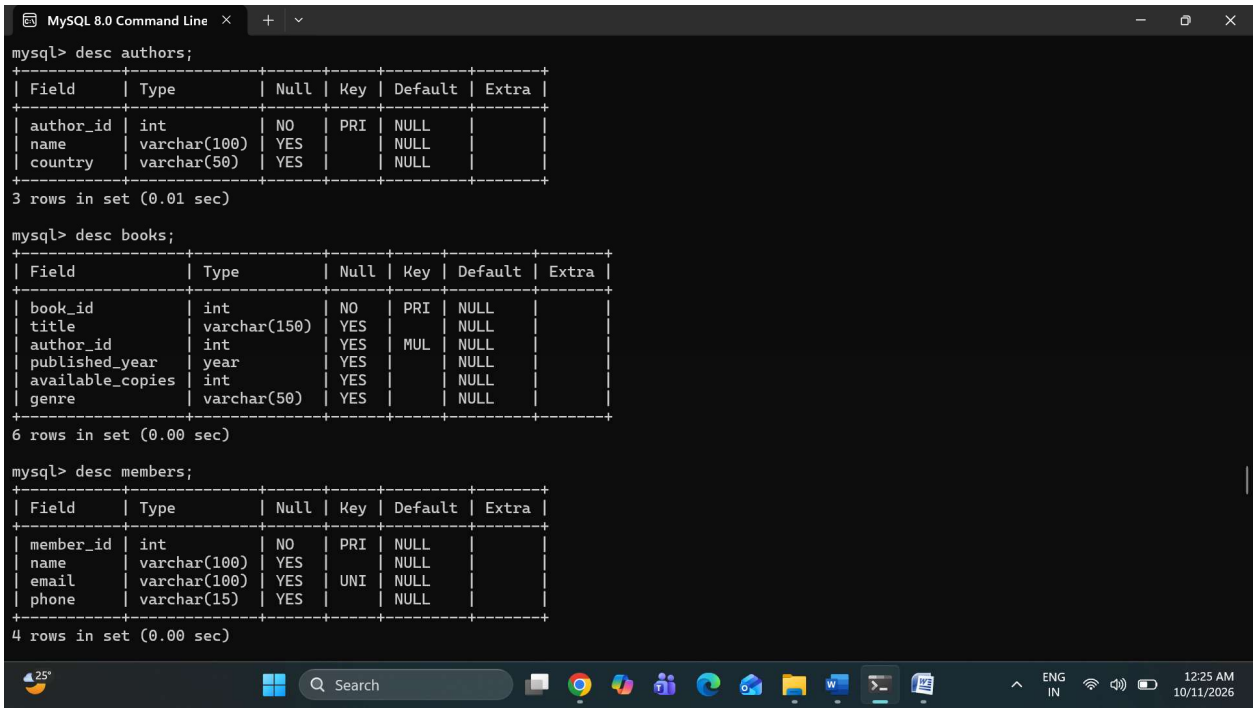
member_id INT PRIMARY KEY,

name VARCHAR(100),

email VARCHAR(100) UNIQUE,

phone VARCHAR(15)

);



Q5. CREATE TABLE borrow_records (

record_id INT PRIMARY KEY,

member_id INT,

book_id INT,

borrow_date DATE,

return_date DATE,

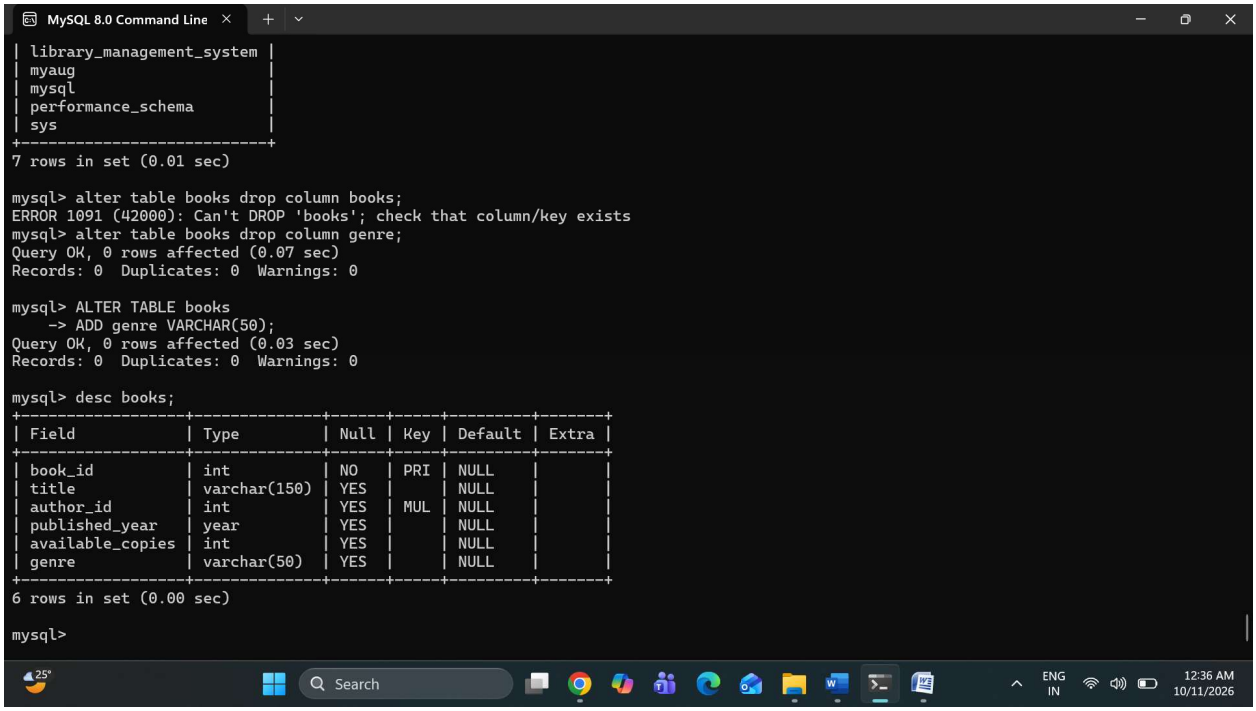
FOREIGN KEY (member_id) REFERENCES Members(member_id),

FOREIGN KEY (book_id) REFERENCES Books(book_id)

);

```
mysql> desc borrow_records;
+--------------+------+------+-----+---------+-------+
| Field        | Type | Null | Key | Default | Extra |
+--------------+------+------+-----+---------+-------+
| record_id    | int  | NO   | PRI | NULL    |       |
| member_id    | int  | YES  | MUL | NULL    |       |
| book_id      | int  | YES  | MUL | NULL    |       |
| borrow_date  | date | YES  |     | NULL    |       |
| return_date  | date | YES  |     | NULL    |       |
+--------------+------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

Q6.



Q7. DROP TABLE Borrow_Records;

Q8. INSERT INTO authors (author_id, name, country)

VALUES

(1, 'William Shakespeare', 'England'),

(2, 'Chetan Bhagat', 'India'),

(3, 'R.K. Narayan', 'India');

Q9. INSERT INTO books (book_id, title, author_id, published_year, available_copies)

VALUES

(1, 'dotnet', 1, 2024, 4),

(2, 'Mysql', 2, 2023, 2),

(3, 'Java', 2, 2026, 4),

(4, 'Database Systems', 1, 2022, 2),

(5, 'Oopj', 1, 2026, 3);

Q10.

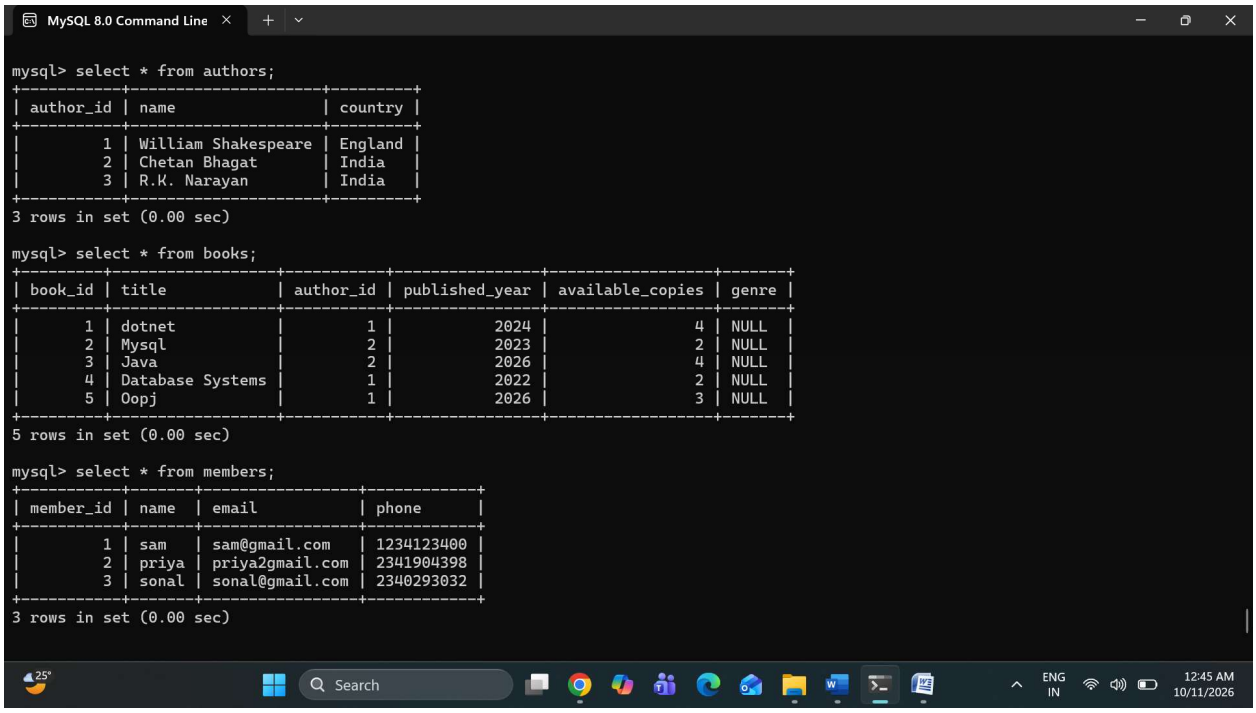
INSERT INTO members (member_id, name, email, phone)

VALUES

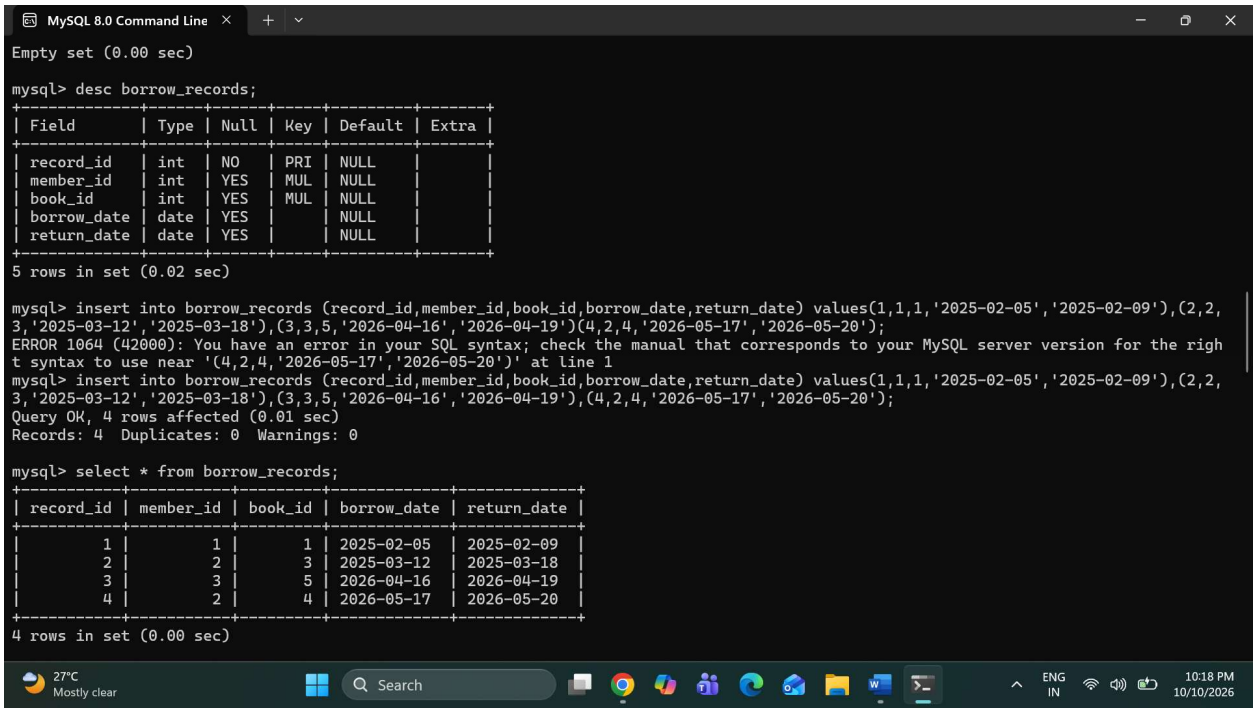
(1, 'sam', 'sam@gmail.com', '1234123400'),

(2, 'priya', 'priya2gmail.com', '2341904398'),

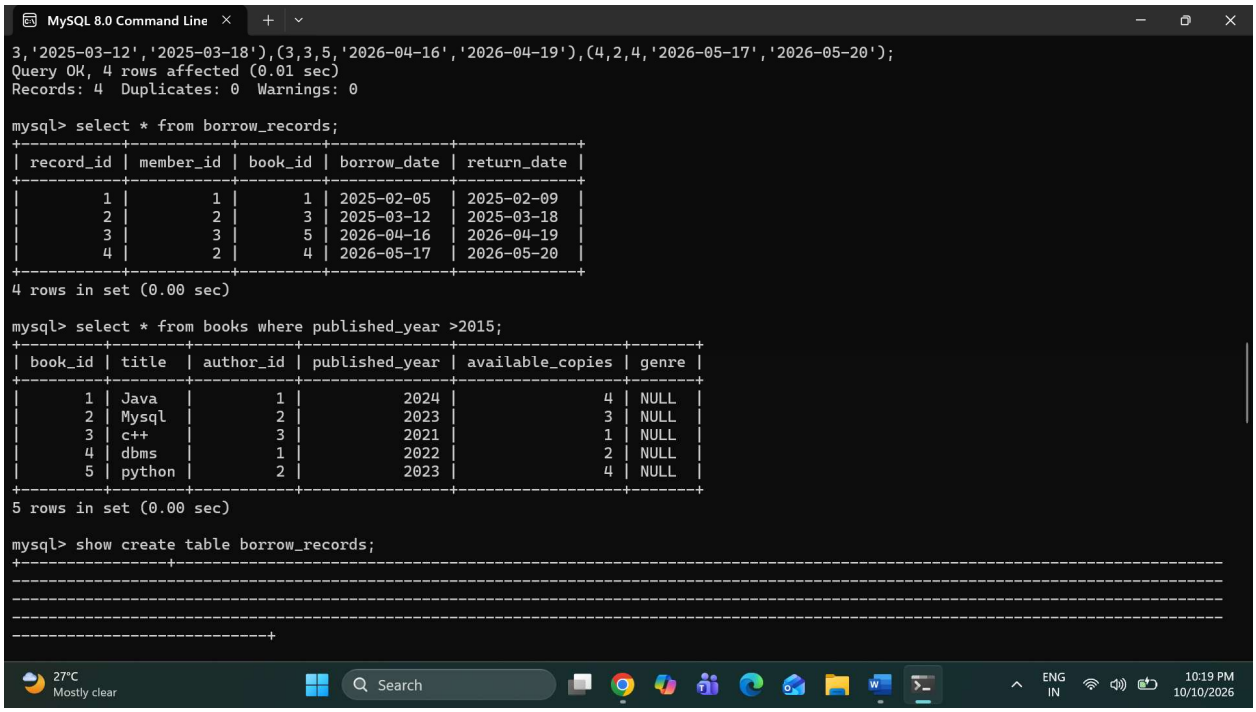
(3, 'sonal', 'sonal@gmail.com', '2340293032');


MySQL 8.0 Command Line
mysql> select * from authors;
+-----------+---------------------+---------+
| author_id | name | country |
+-----------+---------------------+---------+
1	William Shakespeare	England
2	Chetan Bhagat	India
3	R.K. Narayan	India
+-----------+---------------------+---------+
3 rows in set (0.00 sec)
mysql> select * from books;
+---------+------------------+-----------+----------------+------------------+-------+
| book_id | title | author_id | published_year | available_copies | genre |
+---------+------------------+-----------+----------------+------------------+-------+
1	dotnet	1	2024	4	NULL
2	Mysql	2	2023	2	NULL
3	Java	2	2026	4	NULL
4	Database Systems	1	2022	2	NULL
5	Oopj	1	2026	3	NULL
+---------+------------------+-----------+----------------+------------------+-------+
5 rows in set (0.00 sec)
mysql> select * from members;
+-----------+-------+-----------------+------------+
| member_id | name | email | phone |
+-----------+-------+-----------------+------------+
1	sam	sam@gmail.com	1234123400
2	priya	priya2gmail.com	2341904398
3	sonal	sonal@gmail.com	2340293032
+-----------+-------+-----------------+------------+
3 rows in set (0.00 sec)
Search
ENG
IN
12:45 AM
10/11/2026

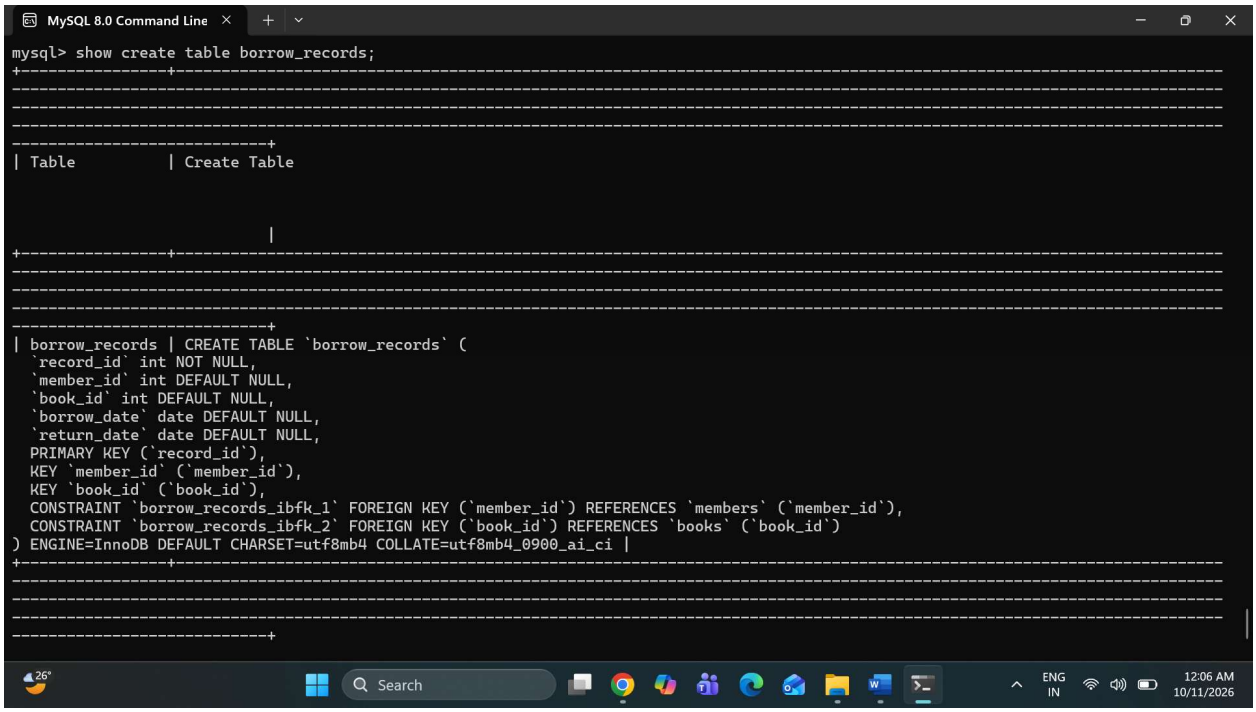
Q11.


MySQL 8.0 Command Line
Empty set (0.00 sec)
mysql> desc borrow_records;
+-------------+------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
record_id	int	NO	PRI	NULL	
member_id	int	YES	MUL	NULL	
book_id	int	YES	MUL	NULL	
borrow_date	date	YES		NULL	
return_date	date	YES		NULL	
+-------------+------+------+-----+---------+-------+
5 rows in set (0.02 sec)
mysql> insert into borrow_records (record_id,member_id,book_id,borrow_date,return_date) values(1,1,1,'2025-02-05','2025-02-09'),(2,2,
3,'2025-03-12','2025-03-18'),(3,3,5,'2026-04-16','2026-04-19')(4,2,4,'2026-05-17','2026-05-20');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near '(4,2,4,'2026-05-17','2026-05-20')' at line 1
mysql> insert into borrow_records (record_id,member_id,book_id,borrow_date,return_date) values(1,1,1,'2025-02-05','2025-02-09'),(2,2,
3,'2025-03-12','2025-03-18'),(3,3,5,'2026-04-16','2026-04-19'),(4,2,4,'2026-05-17','2026-05-20');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
mysql> select * from borrow_records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
1	1	1	2025-02-05	2025-02-09
2	2	3	2025-03-12	2025-03-18
3	3	5	2026-04-16	2026-04-19
4	2	4	2026-05-17	2026-05-20
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)
27°C
Mostly clear
Search
ENG
IN
10:18 PM
10/10/2026

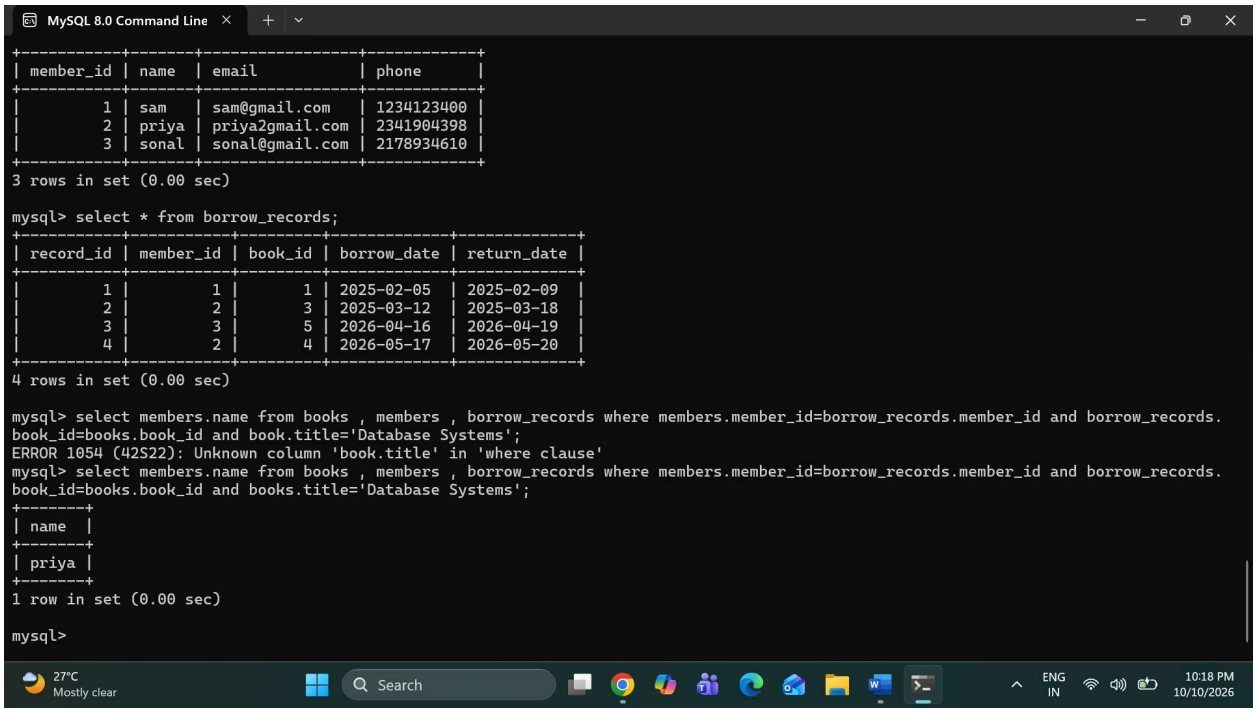
Q12.


MySQL 8.0 Command Line
3,'2025-03-12','2025-03-18'),(3,3,5,'2026-04-16','2026-04-19'),(4,2,4,'2026-05-17','2026-05-20');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
mysql> select * from borrow_records;
record_id	member_id	book_id	borrow_date	return_date
1	1	1	2025-02-05	2025-02-09
2	2	3	2025-03-12	2025-03-18
3	3	5	2026-04-16	2026-04-19
4	2	4	2026-05-17	2026-05-20
4 rows in set (0.00 sec)
mysql> select * from books where published_year >2015;
book_id	title	author_id	published_year	available_copies	genre
1	Java	1	2024	4	NULL
2	Mysql	2	2023	3	NULL
3	c++	3	2021	1	NULL
4	dbms	1	2022	2	NULL
5	python	2	2023	4	NULL
5 rows in set (0.00 sec)
mysql> show create table borrow_records;
27°C
Mostly clear
Search
ENG
IN
10:19 PM
10/10/2026

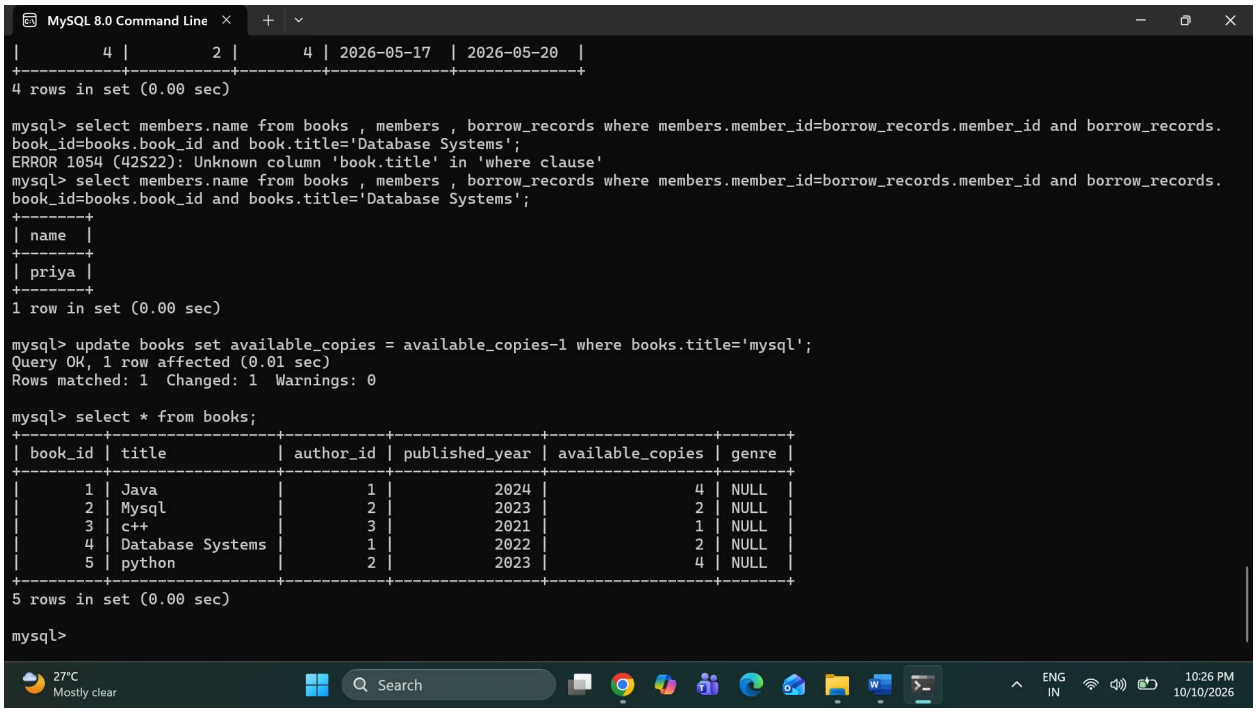
Q13.


MySQL 8.0 Command Line
mysql> show create table borrow_records;
| Table | Create Table
| borrow_records | CREATE TABLE `borrow_records` (
`record_id` int NOT NULL,
`member_id` int DEFAULT NULL,
`book_id` int DEFAULT NULL,
`borrow_date` date DEFAULT NULL,
`return_date` date DEFAULT NULL,
PRIMARY KEY (`record_id`),
KEY `member_id` (`member_id`),
KEY `book_id` (`book_id`),
CONSTRAINT `borrow_records_ibfk_1` FOREIGN KEY (`member_id`) REFERENCES `members` (`member_id`),
CONSTRAINT `borrow_records_ibfk_2` FOREIGN KEY (`book_id`) REFERENCES `books` (`book_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
26°
Search
ENG
IN
12:06 AM
10/11/2026

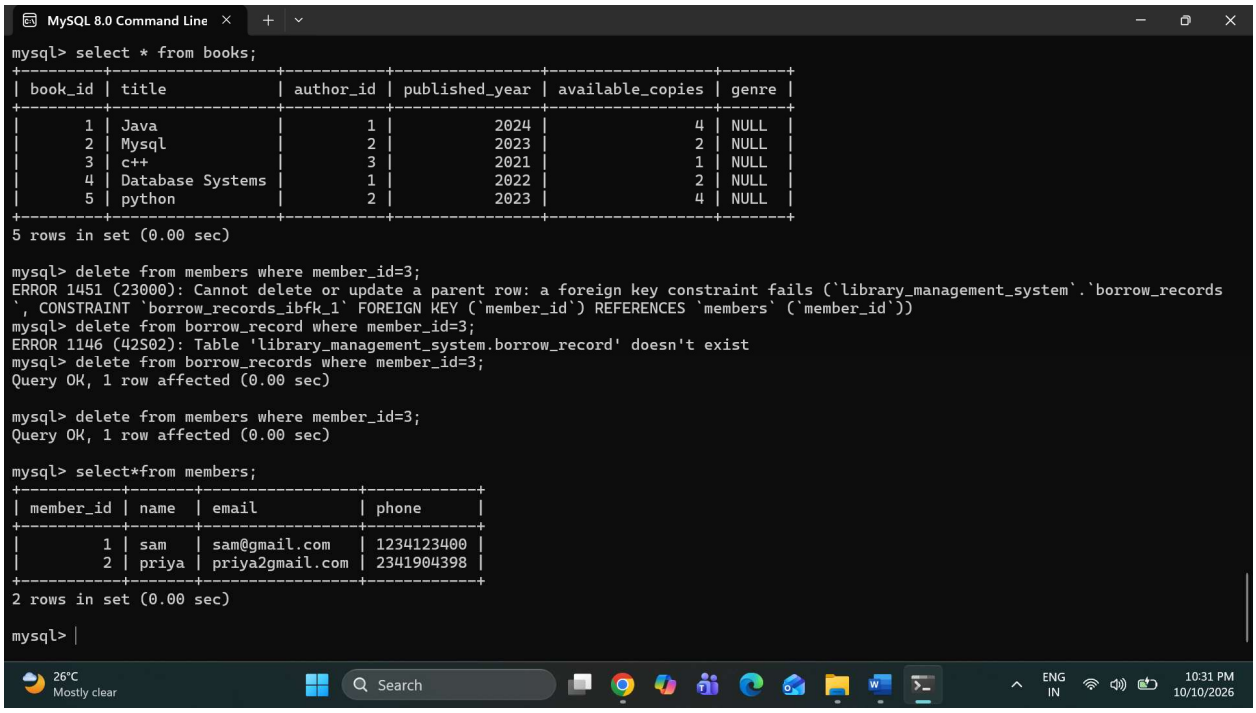
Q14.


MySQL 8.0 Command Line
+-----------+-------+-----------------+------------+
| member_id | name | email | phone |
+-----------+-------+-----------------+------------+
1	sam	sam@gmail.com	1234123400
2	priya	priya2gmail.com	2341904398
3	sonal	sonal@gmail.com	2178934610
+-----------+-------+-----------------+------------+
3 rows in set (0.00 sec)
mysql> select * from borrow_records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
1	1	1	2025-02-05	2025-02-09
2	2	3	2025-03-12	2025-03-18
3	3	5	2026-04-16	2026-04-19
4	2	4	2026-05-17	2026-05-20
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)
mysql> select members.name from books , members , borrow_records where members.member_id=borrow_records.member_id and borrow_records.
book_id=books.book_id and book.title='Database Systems';
ERROR 1054 (42S22): Unknown column 'book.title' in 'where clause'
mysql> select members.name from books , members , borrow_records where members.member_id=borrow_records.member_id and borrow_records.
book_id=books.book_id and books.title='Database Systems';
+-------+
| name |
+-------+
| priya |
+-------+
1 row in set (0.00 sec)
mysql>
27°C
Mostly clear
Search
ENG
IN
10:18 PM
10/10/2026

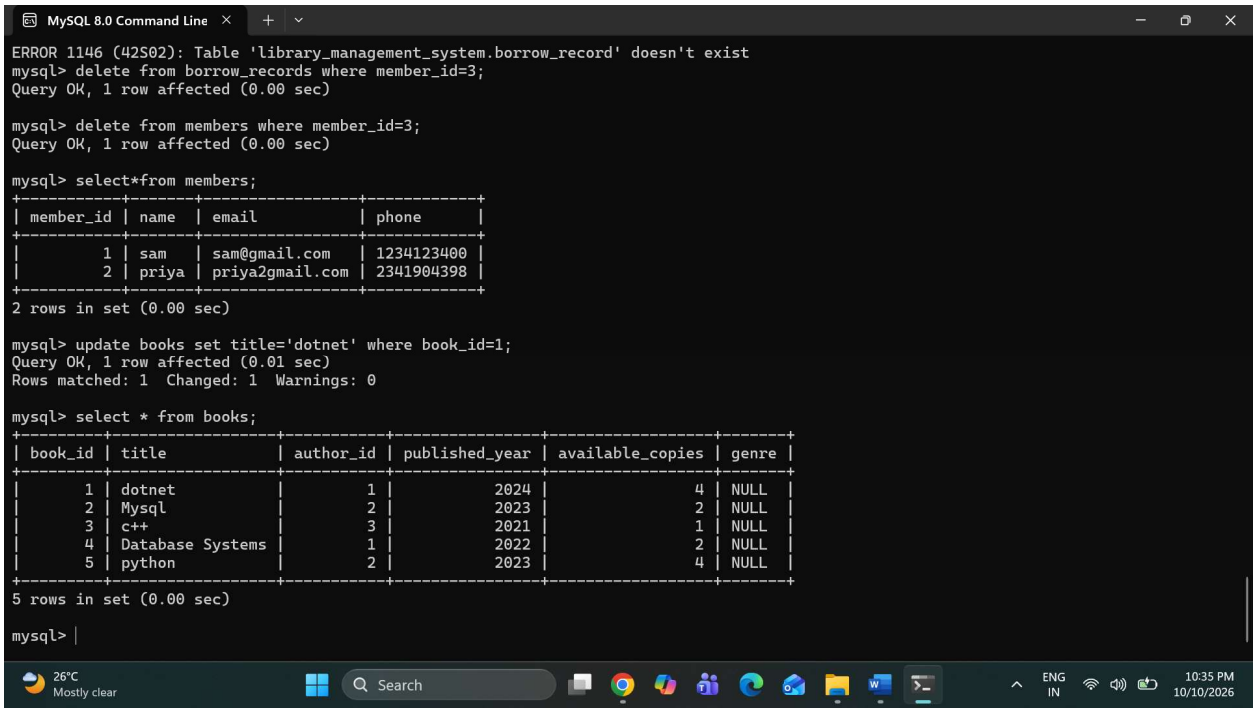
Q15.


MySQL 8.0 Command Line
| 4 | 2 | 4 | 2026-05-17 | 2026-05-20 |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)
mysql> select members.name from books , members , borrow_records where members.member_id=borrow_records.member_id and borrow_records.
book_id=books.book_id and book.title='Database Systems';
ERROR 1054 (42S22): Unknown column 'book.title' in 'where clause'
mysql> select members.name from books , members , borrow_records where members.member_id=borrow_records.member_id and borrow_records.
book_id=books.book_id and books.title='Database Systems';
+-------+
| name |
+-------+
| priya |
+-------+
1 row in set (0.00 sec)
mysql> update books set available_copies = available_copies-1 where books.title='mysql';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
mysql> select * from books;
+---------+------------------+-----------+----------------+------------------+-------+
| book_id | title | author_id | published_year | available_copies | genre |
+---------+------------------+-----------+----------------+------------------+-------+
1	Java	1	2024	4	NULL
2	Mysql	2	2023	2	NULL
3	c++	3	2021	1	NULL
4	Database Systems	1	2022	2	NULL
5	python	2	2023	4	NULL
+---------+------------------+-----------+----------------+------------------+-------+
5 rows in set (0.00 sec)
mysql>
27°C
Mostly clear
Search
ENG
IN
10:26 PM
10/10/2026

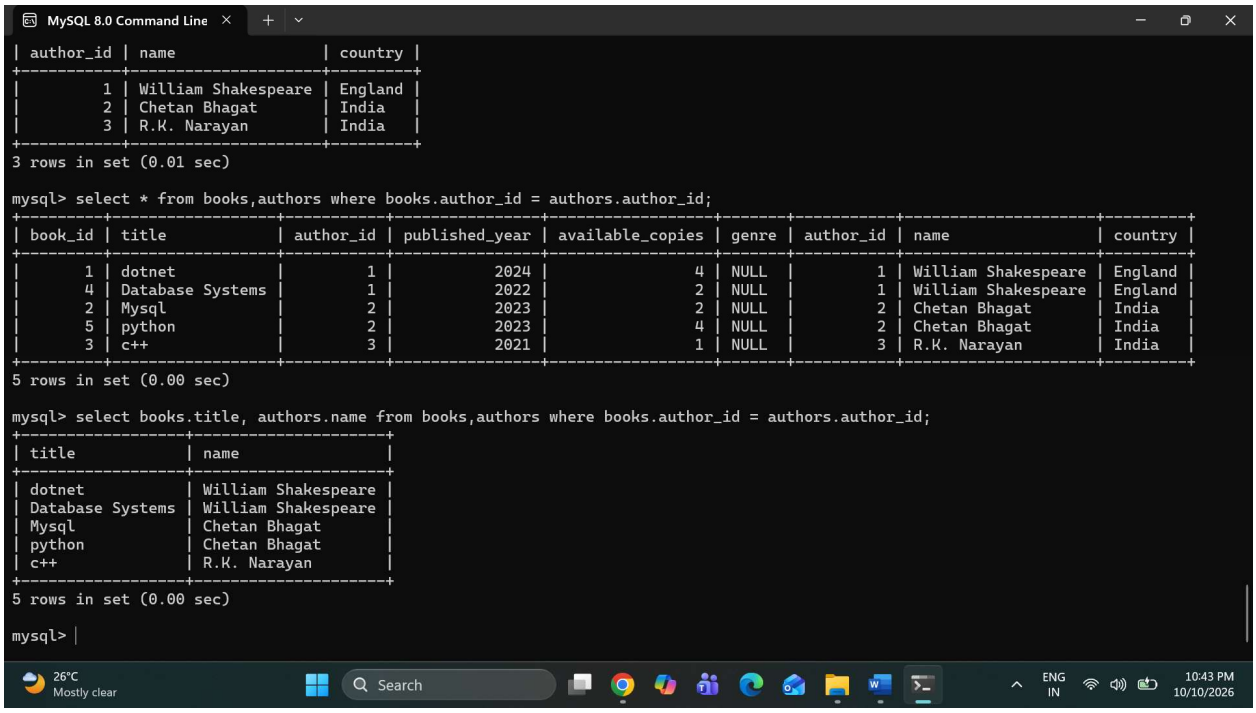
Q16.


MySQL 8.0 Command Line
mysql> select * from books;
+---------+------------------+-----------+----------------+------------------+-------+
| book_id | title | author_id | published_year | available_copies | genre |
+---------+------------------+-----------+----------------+------------------+-------+
1	Java	1	2024	4	NULL
2	Mysql	2	2023	2	NULL
3	c++	3	2021	1	NULL
4	Database Systems	1	2022	2	NULL
5	python	2	2023	4	NULL
+---------+------------------+-----------+----------------+------------------+-------+
5 rows in set (0.00 sec)
mysql> delete from members where member_id=3;
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`library_management_system`.`borrow_records
`, CONSTRAINT `borrow_records_ibfk_1` FOREIGN KEY (`member_id`) REFERENCES `members` (`member_id`))
mysql> delete from borrow_record where member_id=3;
ERROR 1146 (42S02): Table 'library_management_system.borrow_record' doesn't exist
mysql> delete from borrow_records where member_id=3;
Query OK, 1 row affected (0.00 sec)
mysql> delete from members where member_id=3;
Query OK, 1 row affected (0.00 sec)
mysql> select*from members;
+-----------+-------+-----------------+------------+
| member_id | name | email | phone |
+-----------+-------+-----------------+------------+
| 1 | sam | sam@gmail.com | 1234123400 |
| 2 | priya | priya2gmail.com | 2341904398 |
+-----------+-------+-----------------+------------+
2 rows in set (0.00 sec)
mysql>
26°C Mostly clear
Search
ENG IN
10:31 PM 10/10/2026

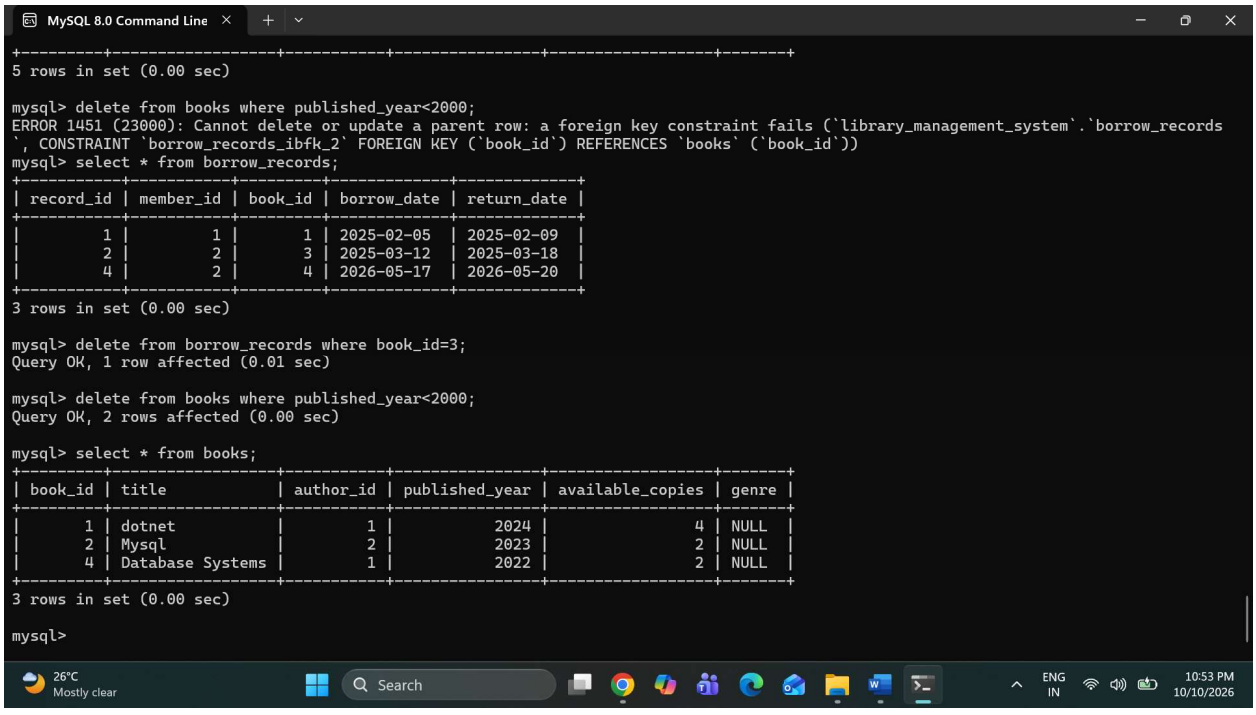
Q17.


MySQL 8.0 Command Line
ERROR 1146 (42S02): Table 'library_management_system.borrow_record' doesn't exist
mysql> delete from borrow_records where member_id=3;
Query OK, 1 row affected (0.00 sec)
mysql> delete from members where member_id=3;
Query OK, 1 row affected (0.00 sec)
mysql> select*from members;
+-----------+-------+-----------------+------------+
| member_id | name | email | phone |
+-----------+-------+-----------------+------------+
| 1 | sam | sam@gmail.com | 1234123400 |
| 2 | priya | priya2gmail.com | 2341904398 |
+-----------+-------+-----------------+------------+
2 rows in set (0.00 sec)
mysql> update books set title='dotnet' where book_id=1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
mysql> select * from books;
+---------+------------------+-----------+----------------+------------------+-------+
| book_id | title | author_id | published_year | available_copies | genre |
+---------+------------------+-----------+----------------+------------------+-------+
1	dotnet	1	2024	4	NULL
2	Mysql	2	2023	2	NULL
3	c++	3	2021	1	NULL
4	Database Systems	1	2022	2	NULL
5	python	2	2023	4	NULL
+---------+------------------+-----------+----------------+------------------+-------+
5 rows in set (0.00 sec)
mysql>
26°C Mostly clear
Search
ENG IN
10:35 PM 10/10/2026

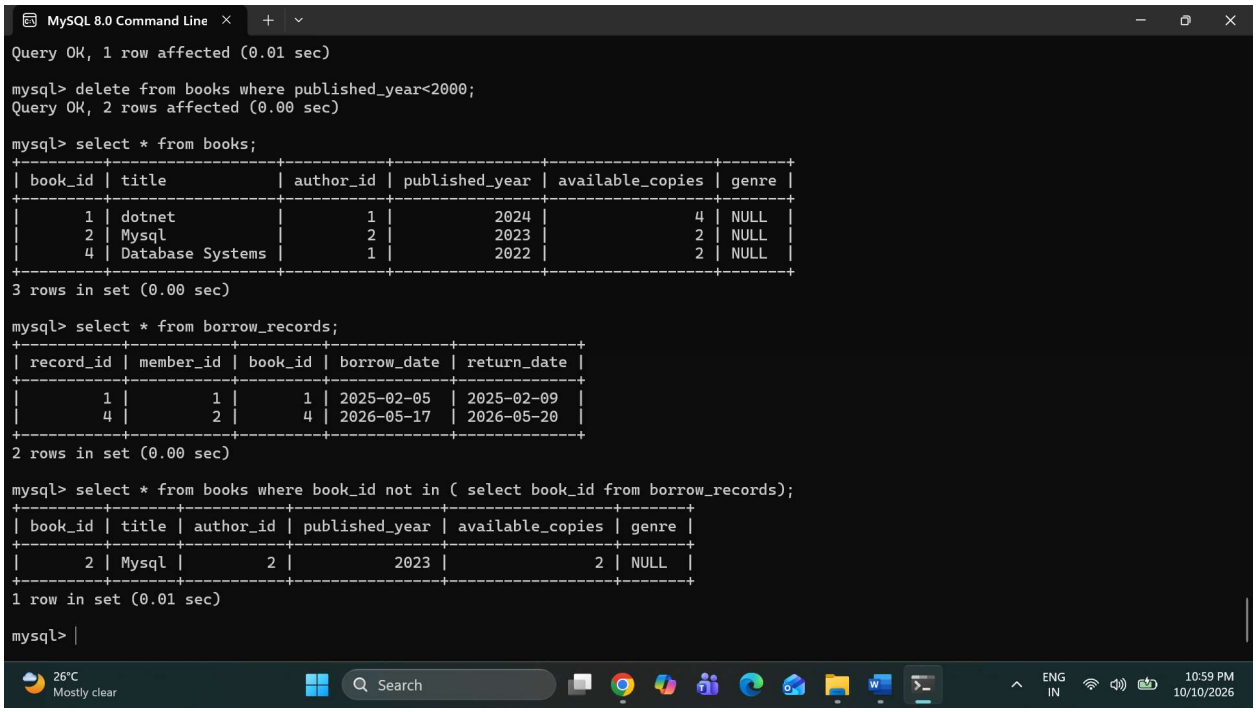
Q18.


MySQL 8.0 Command Line
author_id	name	country
1	William Shakespeare	England
2	Chetan Bhagat	India
3	R.K. Narayan	India
3 rows in set (0.01 sec)
mysql> select * from books,authors where books.author_id = authors.author_id;
book_id	title	author_id	published_year	available_copies	genre	author_id	name	country
1	dotnet	1	2024	4	NULL	1	William Shakespeare	England
4	Database Systems	1	2022	2	NULL	1	William Shakespeare	England
2	Mysql	2	2023	2	NULL	2	Chetan Bhagat	India
5	python	2	2023	4	NULL	2	Chetan Bhagat	India
3	c++	3	2021	1	NULL	3	R.K. Narayan	India
5 rows in set (0.00 sec)
mysql> select books.title, authors.name from books,authors where books.author_id = authors.author_id;
title	name
dotnet	William Shakespeare
Database Systems	William Shakespeare
Mysql	Chetan Bhagat
python	Chetan Bhagat
c++	R.K. Narayan
5 rows in set (0.00 sec)
mysql>
26°C Mostly clear
Search
ENG IN
10:43 PM 10/10/2026

Q19.


MySQL 8.0 Command Line
5 rows in set (0.00 sec)
mysql> delete from books where published_year<2000;
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`library_management_system`.`borrow_records`, CONSTRAINT `borrow_records_ibfk_2` FOREIGN KEY (`book_id`) REFERENCES `books` (`book_id`))
mysql> select * from borrow_records;
record_id	member_id	book_id	borrow_date	return_date
1	1	1	2025-02-05	2025-02-09
2	2	3	2025-03-12	2025-03-18
4	2	4	2026-05-17	2026-05-20
3 rows in set (0.00 sec)
mysql> delete from borrow_records where book_id=3;
Query OK, 1 row affected (0.01 sec)
mysql> delete from books where published_year<2000;
Query OK, 2 rows affected (0.00 sec)
mysql> select * from books;
book_id	title	author_id	published_year	available_copies	genre
1	dotnet	1	2024	4	NULL
2	Mysql	2	2023	2	NULL
4	Database Systems	1	2022	2	NULL
3 rows in set (0.00 sec)
mysql>
26°C Mostly clear
Search
ENG IN
10:53 PM 10/10/2026

Q20.


MySQL 8.0 Command Line
Query OK, 1 row affected (0.01 sec)
mysql> delete from books where published_year<2000;
Query OK, 2 rows affected (0.00 sec)
mysql> select * from books;
+---------+------------------+-----------+----------------+------------------+-------+
| book_id | title | author_id | published_year | available_copies | genre |
+---------+------------------+-----------+----------------+------------------+-------+
1	dotnet	1	2024	4	NULL
2	Mysql	2	2023	2	NULL
4	Database Systems	1	2022	2	NULL
+---------+------------------+-----------+----------------+------------------+-------+
3 rows in set (0.00 sec)
mysql> select * from borrow_records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
| 1 | 1 | 1 | 2025-02-05 | 2025-02-09 |
| 4 | 2 | 4 | 2026-05-17 | 2026-05-20 |
+-----------+-----------+---------+-------------+-------------+
2 rows in set (0.00 sec)
mysql> select * from books where book_id not in (select book_id from borrow_records);
+---------+-------+-----------+----------------+------------------+-------+
| book_id | title | author_id | published_year | available_copies | genre |
+---------+-------+-----------+----------------+------------------+-------+
| 2 | Mysql | 2 | 2023 | 2 | NULL |
+---------+-------+-----------+----------------+------------------+-------+
1 row in set (0.01 sec)
mysql>
26°C
Mostly clear
Search
ENG
IN
10:59 PM
10/10/2026